A Parameter-Free Prediction of the Solar Mixing Angle from Cubic-Group Flavor Structure, Tested by JUNO

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Date: April 2026
Status: DRAFT PRE-PRINT
Classification: Theoretical Physics / High Energy Physics / Neutrino Phenomenology

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Abstract

We derive the parameter-free prediction $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2) = 0.3080$ from cubic-group flavor structure, with no parameter fit to the value of $\sin^2\theta_{12}$. The first JUNO measurement of reactor neutrino oscillations reports $\sin^2\theta_{12} = 0.3092 \pm 0.0087$, with the post-JUNO global fit tightening this to $\sin^2\theta_{12} = 0.3085 \pm 0.0073$, matching the prediction at 0.07σ. The same construction gives $\sin^2\theta_{13} = 4/(18\pi^2) = 0.02252$ and $\sin^2\theta_{23} = 1/2 + 1/(2\pi^2) = 0.5507$, each consistent with NuFIT 6.0 [14] within 1.1σ, and forces the exact sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ — a clean discriminator from the TM1 and TM2 patterns currently analyzed against JUNO. The angle predictions follow from a discrete μ-τ parity together with one cubic-group structural relation between the U-even and U-odd components of the second-order perturbation. The Cabibbo angle $\lambda = 1/(\pi\sqrt{2}) = 0.22508$, set by the cubic Brillouin-zone geometry, sets the perturbation scale and matches the observed value to 0.04%. JUNO's design-lifetime precision of ± 0.0014 will test the prediction at sub-percent level.

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1. Introduction

1.1 The result

The first JUNO measurement of reactor neutrino oscillations [7] reports

$$\sin^2\theta_{12} = \mathbf{0.3092 \pm 0.0087}$$

from 59.1 days of data. The post-JUNO global fit of Capozzi *et al.* [8] tightens this to

$$\sin^2\theta_{12} = \mathbf{0.3085 \pm 0.0073}.$$

We show that taking the cubic point group O — the rotation symmetry group of three-dimensional space — as the underlying flavor structure of the lepton sector yields the prediction

$$\sin^2\theta_{12} = \mathbf{1/3 - 1/(4\pi^2) = 0.30798}$$

matching the post-JUNO global fit at $\mathbf{0.}07\sigma$ with no parameter fit to $\sin^2\theta_{12}$. The same construction predicts $\sin^2\theta_{13}$ and $\sin^2\theta_{23}$ within 1.1σ of the NuFIT 6.0 best fits [14] and forces the exact sum rule

$$\mathbf{2}\sin^2\theta_{12} + \sin^2\theta_{23} = \mathbf{7/6}$$

which distinguishes the present prediction from the TM1, TM2, and TM3 column-preservation patterns at JUNO precision. Both inputs to $\sin^2\theta_{12}$ are independently motivated: the zeroth-order value $1/3$ is the A_4 TBM result, and the perturbation scale $\lambda^2 = 1/(2\pi^2)$ is the squared Cabibbo angle, fixed independently by the cubic Brillouin geometry (matching the observed λ to 0.04%).

1.2 Where the prediction comes from

The structure is constrained by two ingredients. First, the perturbation scale $\lambda^2 = 1/(2\pi^2)$ is not a fitted parameter: it is the squared Cabibbo angle, fixed by a single cubic Brillouin-zone distance (Sec. 2.2). Numerically $\lambda = 1/(\pi\sqrt{2})$ $= 0.22508$, agreeing with the observed Cabibbo value 0.22500 ± 0.00067 [13] to 0.04%. Second, the deviations from tribimaximal (TBM) mixing at $O(\lambda^2)$ are controlled by a discrete μ-τ parity (the U-parity of $S_4 \supset A_4$) together with one cubic-group structural relation between the U-even and U-odd components of the second-order perturbation (Sec. 3.4). The structural relation fixes the coefficient of the $1/(4\pi^2)$ correction; without further inputs, the angle predictions are

- $\sin^2\theta_{12} = \mathbf{1/3 - 1/(}4\pi^2\mathbf{) = 0.30798}$
- $\sin^2\theta_{13} = \mathbf{4/(}18\pi^2\mathbf{) = 0.02252}$
- $\sin^2\theta_{23} = \mathbf{1/2 + 1/(}2\pi^2\mathbf{) = 0.55066}$

The 0.07σ match with the post-JUNO global fit therefore tests structural relations rather than fitted parameters.

1.3 The empirical context

The cubic-group structure used here is not a postulated flavor symmetry. It is the rotation symmetry group of three-dimensional space, realized in a lattice-based framework [18] in which the simple cubic Bravais lattice in d=3 is shown to be the unique three-dimensional Bravais lattice consistent with the SU(3) $\times$ SU(2) $\times$ U(1) gauge structure and the observed Cabibbo angle (Theorem 7b). The same cubic structure that produces the $1/(4\pi^2)$ correction here also produces a number of related quantitative predictions across the flavor sector, all empirically tested:

Observable	Cubic-group prediction	Observed	Match
Cabibbo angle λ	$1/(\pi\sqrt{2}) = 0.22508$	0.22500 ± 0.00067 [13]	0.04%

Observable	Cubic-group prediction	Observed	Match
Koide angle θ_0 (charged leptons)	$C_2/d^2 = 2/9$	0.222204 ± 0.000010 [13]	0.02%
Solar mixing $\sin^2\theta_{12}$ (this work)	$1/3 - 1/(4\pi^2) = 0.30798$	0.3085 ± 0.0073 [8]	0.07σ
PMNS sum rule	$2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$	1.178 ± 0.020	0.6σ

The first two entries are derived in [18, §§7.1–7.2]; the Cabibbo derivation is unconditional structural (the chirality structure of the taste-changing vertex is established in the staggered fermion literature [18, §7.1, citing Mason et al.]), and the empirical match to 0.04% supports this status. The Koide angle is unconditional structural at first-principles. The third and fourth entries are the present work, classified at the layered level (Layer 1 form combined with the Layer 2(a)-tractable Condition 2 coefficient) per [18, §8.3]; see Sec. 6.3 for full discussion. This cumulative empirical track record is what distinguishes the cubic-group structure from postulated flavor symmetries: the same parameter $\lambda^2 = 1/(2\pi^2)$ and the same projection factor $A^2 = 2/3$ enter multiple observables at sub-percent precision, which is testable structure rather than symmetry assumption.

1.4 Position in the broader literature

The lattice-based framework in which the cubic-group structure is realized [18] is one of several recent constructions in which observer-essentiality plays a foundational role. Chandrasekaran, Longo, Penington, and Witten [19] established that incorporating an observer promotes the von Neumann algebra of QFT observables in a gravitational subregion from type III to type II, with a well-defined entropy. Maldacena [20] showed that an observer-with-clock cancels the unphysical phase of the de Sitter sphere partition function. Harlow, Usatyuk, and Zhao [21] argued that the closed-universe Hilbert space dimension is determined by the observer's degrees of freedom. Slagle and Preskill [22] constructed boundary quantum mechanics from a classical bulk lattice model with stochastic dynamics, demonstrating the logical possibility of QM-from-classical at the cost of an exponentially long extra spatial dimension. The framework underlying this paper differs from these constructions in being deterministic at the substratum level (no stochastic dynamics, no extra spatial dimension), and in producing the cubic-group flavor structure used here as a structural consequence rather than an input. The convergence with the post-2022 mainstream observer-emergent literature on the foundational observer-essentiality move is supportive context for the present prediction; the prediction itself stands or falls on its empirical match with JUNO, which is the focus of this paper.

The remainder of this paper develops the technical content. Section 2 sets up the cubic-group structure and derives the perturbation scale. Section 3 computes the deviations from TBM at $O(\lambda^2)$ and derives the sum rule. Section 4 presents the predictions and compares them with JUNO, the post-JUNO global fit, and competing TM1/TM2 patterns. Section 5 covers the remaining angles. Section 6 addresses robustness. Section 7 discusses falsifiability under JUNO's projected long-term precision. Section 8 concludes.

2. Cubic-group flavor structure

2.1 The cubic group O and its A_4 subgroup

The cubic point group O is the rotation symmetry group of three-dimensional space, the unique finite point group with a six-element generator set $\{\pm\hat{e}_1, \pm\hat{e}_2, \pm\hat{e}_3\}$ — six minimal-length translations corresponding to forward and backward motion along three orthogonal axes. In the lattice realization of the OI framework, this six-element set arises from coupling-degree minimization (K = 2d = 6 for d = 3), and the multiplicities (3, 2, 1) under the action of O on these six directions match the structure of one Standard Model fermion generation (quark triplet, lepton doublet, singlet). Three SM generations are placed on the T_1 representation of the cubic group O acting on the three Cartesian axes — a forced consequence of [18, §4.6 Theorem 7] (cubic decomposition $6 = T_1 \oplus E \oplus A_1$), [18, §4.6 Theorem 7b] (Bravais-lattice uniqueness — SC is the unique 3D Bravais lattice compatible with the SM gauge group at the commutant level and with the observed Cabibbo angle), [18, §4.7 Theorem 10] (cubic symmetry forces three degenerate triplet sectors), [18, §4.7.1.2] (link-carrier construction placing matter on links with cross-charged representations admissible), and [18, Theorem 15] (anomaly cancellation fixing hypercharges uniquely). The detailed argument from coupling-degree minimization through the (3, 2, 1) decomposition is given in [18, §§4.5–4.7].

The full octahedral rotation group O has order 24 and is isomorphic to S_4. Its irreducible representations decompose as

$$\mathbf{irreps(O)} = A_1 \oplus A_2 \oplus \mathbf{E} \oplus T_1 \oplus T_2$$

of dimensions 1, 1, 2, 3, 3. The subgroup $A_4 \subset$ O keeps $\{A_1, A_2, \mathrm{E}, T_1\}$ but identifies A_2 with A_1, giving the standard A_4 content A ⊕ A' ⊕ A'' ⊕ T of dimensions 1, 1, 1, 3.

We take T_1 as the generation triplet. Realize it by the three Cartesian unit vectors $\{e_1, e_2, e_3\}$ acted on by the 24 rotations of O. The democratic vector

$$\hat{h} = (\mathbf{1}/\sqrt{3})(e_1 + e_2 + e_3)$$

is the only direction fixed by the $\mathbb{Z}_3$ cyclic subgroup. The plane perpendicular to $\hat{h}$ carries the doublet E. This is the geometric origin of TBM: the second column of U_TBM is exactly $\hat{h}$, and the first and third columns lie in the plane perpendicular to it, oriented by μ-τ exchange.

2.2 The Cabibbo scale from cubic Brillouin geometry

We identify the perturbation parameter λ with the Cabibbo angle. In a lattice realization of the cubic-group flavor structure, the three generations occupy the three T_1 corners of the cubic Brillouin zone at momenta X_j = $\pi \cdot e$_j (j = 1, 2, 3). The inter-generation distance in momentum space is

$$|\mathbf{X_i - X_j}| = \pi\sqrt{2}\ \mathbf{(i \neq j)}$$

a fixed geometric constant of the cubic lattice. The leading-order inter-generation mixing matrix element is the continuum fermion propagator at this momentum, |M_ij| = |S(X_j − X_i)| = 1/|X_j − X_i|. The 1/|q| scaling (rather than $1/|q|^2$) is structural: in the massless limit, the chirality-preserving component of the fermion propagator at momentum q reduces to S(q) = $\gamma \cdot q/q^2$, and the taste-changing vertex's Dirac trace contracts this to a magnitude of order 1/|q| rather than the full propagator's $1/|q|^2$. The full derivation, including the chirality verification of the taste-changing vertex's spinor structure, is given in [18, §7.1]. Identifying the matrix element with the Cabibbo angle gives

$$\lambda = \mathbf{1/(\pi\sqrt{2}) = 0.22508}$$

matching the observed value λ_obs = 0.22500 ± 0.00067 [13] to 0.04%. Squaring,

$$\lambda^2 = \mathbf{1/(2\pi^2) = 0.05066}$$

sets the overall scale of O(λ^2) corrections. The full derivation of $\lambda = 1/(\pi\sqrt{2})$ from the lattice fermion propagator and the chirality-preservation argument is given in [18, §7.1]. The empirical match to 0.04% is at the level of structural relations rather than fitted parameters.

A second geometric quantity will appear. The angle between any generation axis e_i and the democratic direction $\hat{h}$ satisfies e_i $\cdot\hat{h} = 1/\sqrt{3}$, so $\cos^2(\angle$(e_i, $\hat{h}$)) = 1/3 and $\sin^2(\angle$(e_i, $\hat{h}$)) = 2/3. We identify the Wolfenstein parameter A with this quantity:

$$\mathbf{A} = \sqrt{2/3} = \mathbf{0.8165}$$

agreeing with the PDG value A = 0.826 ± 0.012 at 0.23σ. The combination A^2 = 2/3 enters the inter-generation projection geometry, and A^4 = 4/9 appears directly in the deviation Δ_{13} below.

2.3 The PMNS matrix in the Altarelli-Feruglio basis

We work in the Altarelli-Feruglio (AF) basis [12]. In this basis the neutrino sector realizes TBM exactly (after diagonalization of the seesaw matrix), and all deviations from TBM come from the charged-lepton mass matrix. The PMNS matrix is

$$\mathbf{U_PMNS} = \mathbf{V_}\ell^\dagger \cdot \mathbf{U_TBM}$$

where V_ℓ is the charged-lepton rotation that diagonalizes M_ℓ M_$\ell^\dagger$, and

$$U_{\mathrm{TBM}} = P0$$

Expanding V_ℓ in λ,

$\mathbf{V_\ell = \exp(\lambda A_1 + \lambda^2 A_2 + O(\lambda^3))}$

with A_1, A_2 real antisymmetric 3×3 matrices. The remaining task is to determine A_1 and A_2 from the structural ingredients of the cubic geometry.

3. U-parity grading and the sum rule

3.1 The U-parity generator

In the AF basis, the μ-τ exchange generator U $\in S_4$ acts on the generation indices by exchanging the second and third:

$$U = P1$$

Conjugation by U defines a $\mathbb{Z}_2$ grading on antisymmetric matrices: A is **U-even** if $UAU^{-1} = +$A, **U-odd** if $UAU^{-1} = -$A. With (E_{ij})*{kl}* = δ{ik}δ_{jl} $-\delta$_{il}δ_{jk}, the decomposition is:

- **U-even**: $E_{12} + E_{13}$
- **U-odd**: $E_{12} - E_{13}$, E_{23}

The unique U-odd combination antisymmetric in (μ, τ) is E_{23}; the unique U-odd combination relating the first generation to the second/third is $E_{12} - E_{13}$.

3.2 Determination of A_1

A_1 is fixed by three structural conditions plus the projection geometry of §2.2.

The data require $\sin\theta_{13} = \mathrm{O}(\lambda)$ (since $\theta_{13} \neq 0$), forcing A_1 to contribute to the (1,3) element of V_$\ell^\dagger$ U_TBM. The observation that both $\Delta_{12} \equiv \sin^2\theta_{12} - 1/3$ and $\Delta_{23} \equiv \sin^2\theta_{23} - 1/2$ are $\mathrm{O}(\lambda^2)$ (not $\mathrm{O}(\lambda)$) further constrains $A_1's$ structure. The remaining input is geometric: each end of an inter-generation mixing vertex projects onto the plane perpendicular to $\hat{h}$, contributing a factor sin($\angle$(e_i, $\hat{h}$)) = A from the off-democratic projection (Sec. 2.2). For a single-vertex inter-generation transition, the total projection factor is A^2 (one factor per vertex end). Identifying this projection geometry with the leading $\mathrm{O}(\lambda)$ contribution to U_e3 gives the structural identity

$\sin\theta_{13} = A^2\lambda$

(at $\mathrm{O}(\lambda)$; A_2 contributes only at $\mathrm{O}(\lambda^2)$ and cancels from $|U_e3|^2$ as shown in §3.3 below). This is the structural input — not a derived consequence — that fixes $A_1's$ overall coefficient up to sign.

The result is

$A_1 = (\sqrt{2}/3)(E_{12} - E_{13})$

A_1 is purely U-odd: $UA_1U^{-1} = -A_1$. Geometrically, it is a rotation by angle $A^2 = 2/3$ around the axis $(e_2 + e_3)/\sqrt{2}$, which reproduces $\sin\theta_{13} = A^2\lambda$ as required.

3.3 Deviations to O(λ^2)

Expanding V_ℓ to second order,

$\mathbf{V_\ell = I + \lambda A_1 + \lambda^2(A_2 + 1/2A_1^2) + O(\lambda^3)}$

and writing the most general antisymmetric A_2 as

$A_2 = b_{12}E_{12} + b_{13}E_{13} + b_{23}E_{23}$

we compute $|U_PMNS, ij|^2$ to O(λ^2). The PMNS angles satisfy $\sin^2\theta_{13} = |U_e3|^2$, $\sin^2\theta_{12}\cos^2\theta_{13} = |U_e2|^2$, and $\sin^2\theta_{23}\cos^2\theta_{13} = |U_\mu3|^2$. Direct computation (taking V_ℓ^T = exp(-(λA_1+$\lambda^2 A_2$)) acting on the columns of U_TBM, using the explicit form A_1^2 = diag(-4/9, -2/9, -2/9) + (2/9)($e_2 \otimes e_3 + e_3 \otimes e_2$) for A_1 = $(\sqrt{2}/3)(E_{12} - E_{13})$) gives the deviations from TBM:

- $\Delta_{13} = \mathbf{(4/9)}\lambda^2 = A^4\lambda^2$
- $\Delta_{12} = -\mathbf{(2/3)}(b_{12} + b_{13})\ \lambda^2$
- $\Delta_{23} = b_{23}\lambda^2$

Three observations follow.

(i) Δ_{13} is fully determined by A_1 alone. The A_2 parameters cancel from $|U_e3|^2$ at O(λ^2), so $\sin^2\theta_{13} = (4/9)\lambda^2$ is fixed without reference to A_2. Equivalently, the structural identity $\sin\theta_{13} = A^2\lambda$ is exact at O(λ) and receives no O(λ^2) correction.

(ii) The combination $b_{12} - b_{13}$ does not enter any of the three angles. It is naturally identified with the CP-violating Dirac phase δ_CP, since it is the only remaining free parameter at O(λ^2) that does not enter the angles. The present analysis leaves δ_CP undetermined.

(iii) Δ_{23} depends only on b_{23}, not on the U-even combination. The explicit calculation gives $|U_\mu3|^2 = 1/2 + \lambda^2(b_{23} - 2/9)$ before the $\cos^2\theta_{13}$ dressing; after dressing, $\sin^2\theta_{23} = 1/2 + \lambda^2 b_{23}$ exactly. The U-even combination (b_{12}+b_{13}) does not contribute to Δ_{23} at O(λ^2): the contributions that would enter through the (V_ℓ^T)$_2\nu$ matrix elements all arise at O(λ^3) or higher in the calculation of $|U_\mu3|^2$.

3.4 Sum rule and absolute normalization

The deviations depend on two scalar combinations of A_2: ($b_{12} + b_{13}$) controlling Δ_{12}, and b_{23} controlling Δ_{23}. We fix the normalization with two conditions.

Condition 1 (natural normalization). Take $b_{23} = 1$. This sets A_2 to enter the perturbation series at unit strength relative to λ^2. It is a labeling convention rather than a physical input — fixing the normalization of the $O(\lambda^2)$ expansion before $b_{12} + b_{13}$ is determined.

Condition 2 (cubic-group structural relation). Take

$$b_{23} = \mathbf{(4/3)}(b_{12} + b_{13})$$

which with Condition 1 gives $b_{12} + b_{13} = 3/4$. Unlike Condition 1, this is a substantive physical input: it relates the dominant U-odd coefficient of A_2 to the U-even combination through a fixed numerical factor 4/3.

Equivalence with TBM sum rule preservation. Combining the explicit results $\Delta_{12} = -(2/3)(b_{12}+b_{13})\lambda^2$ and $\Delta_{23} = b_{23}\lambda^2$ from §3.3:

$$2\Delta_{12} + \Delta_{23} = (b_{23} - \mathbf{(4/3)}(b_{12}+b_{13}))\ \lambda^2$$

Hence Cond 2 is *equivalent* to the statement that the TBM sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ — which holds at zeroth order from cubic geometry, since $2(1/\mathrm{d}) + 1/(\mathrm{d}-1) = (3\mathrm{d}-2)/(\mathrm{d}(\mathrm{d}-1)) = 7/6$ for $\mathrm{d} = 3$ — continues to hold at $O(\lambda^2)$. The Wilson-coefficient relation and the angle-level sum rule are the same content under different normalizations.

Structural protection of the sum rule by A_1 alone (Layer 1). With $A_2 = 0$, direct calculation gives $2|U_e2|^2 + |U_\mu3|^2 = 7/6 + \lambda^2[-14/27] + (\cos^2\theta_{13}$ corrections). Dividing by $\cos^2\theta_{13} = 1 - (4/9)\lambda^2$ (which itself follows from $\sin^2\theta_{13} = A^4\lambda^2$, a Layer 1 quantity) precisely cancels the $-14/27$ offset:

$$**(2\sin^2\theta_{12} + \sin^2\theta_{23})|_\{A_2{=}0\} = 7/6 + O(\lambda^4)**$$

— that is, the sum rule is preserved by A_1 alone at all orders up to λ^2. This cancellation is non-trivial and specific to $A_1's$ Layer 1-forced form: the magnitude $(\sqrt{2}/3)$ is fixed by $\sin\theta_{13} = A^2\lambda$ (§3.2), and the U-odd character is fixed by the requirement Δ_{12}, $\Delta_{23} = O(\lambda^2)$ rather than $O(\lambda)$. With those two constraints, the cancellation between the $-14/27$ contribution from A_1^2-induced shifts and the $+14/27$ contribution from the $\cos^2\theta_{13}$ dressing is automatic. The TBM sum rule is therefore *structurally protected* against $A_1's$ perturbation by cubic geometry alone — no substrate input required for this component.

Cond 2 reduces to the question of whether A_2 preserves the sum rule it was protected for under A_1, or breaks it. The answer depends on the substrate's specific b coefficients.

The factor 4/3. Cond 2's specific numerical coefficient admits the structural reading

$$\mathbf{4/3} = 2A^2 = \mathbf{2}\ \sin^2(\angle(\mathbf{e_i},\ \hat{h}))$$

where $A = \sqrt{2/3}$ is the off-democratic projection of any generation axis (Sec. 2.2). The identity $A^2 = (\mathrm{d}-1)/\mathrm{d} = C_2(T_1)/d$ for $\mathrm{d} = 3$ makes this a representation-theoretic quantity rather than a numerical coincidence: it is the

squared off-democratic projection for the vector representation of any rotation group. The factor 4/3 in Condition 2 admits a *partner-count* $\times$ *per-partner projection* reading: the U-even combination $E_{12}+E_{13}$ couples generation e_1 to its two partners $\{e_2,\ e_3\}$, with the per-partner projection nominally A^2. However, the load-bearing claim of this argument is the *operator alignment* — that the substrate's second-order Yukawa structure has a specific T_2 direction matching the partner-count $\times$ projection picture. Operator counting on $T_1 \otimes T_1 \to T_2$ in the supporting framework [18] shows that this T_2 direction is determined by the substrate's specific Yukawa structure rather than forced by representation theory: the natural Higgs-democratic spurion does not produce Cond 2.

Layer status of Cond 2 after this analysis. The decomposition produced above changes the layer status of the prediction. The A_1 component is Layer 1: the sum rule is structurally protected against $A_1's$ perturbation by cubic geometry, with no substrate input. The A_2 component reduces to a sharply specified open question — *why does $b_{23}/(b_{12}+b_{13})$ take exactly the value 4/3 $= 2A^2$?* — which is Layer 2(a)-tractable rather than the prior Layer 2(b) "arbitrary T_2 direction alignment" formulation. The residual question concerns a single numerical ratio with a structural reading rather than a generic direction in T_2-space; substrate calculations targeting this specific ratio admit definite outcomes. The empirical match at 0.07σ on $\sin^2\theta_{12}$ tests this specific structural feature — Cond 2 with its 4/3 coefficient — rather than testing generic Layer 2(b) alignment.

In standard effective field theory, the U-even and dominant U-odd coefficients $b_{12}+b_{13}$ and b_{23} would be independent Wilson coefficients. The structural relation $b_{23} = (4/3)(b_{12}+b_{13})$ is more naturally accommodated in frameworks where quantum mechanics is itself an emergent phenomenon arising from a deeper deterministic structure [15, 16, 17], since EFT coefficients can then be related by substrate-level mechanisms even absent a protecting EFT symmetry. The factorization $4/3 = 2A^2$ is consistent with this picture; whether it follows rigorously from the cubic-lattice realization of [18] is the open task referenced above.

Sharpened open task. In the AF basis (charged leptons diagonal at LO), second-order perturbative diagonalization of M_ℓ M_$\ell^\dagger$ generates contributions to the A_2 coefficients from cross-terms involving A_1. Direct computation shows that the framework's choice $A_1 = (\sqrt{2}/3)(E_{12} - E_{13})$ produces a hierarchy-independent cross-term contribution b_{23}^cross $= -1/9$ in the m_$e^2 \to 0$ limit, while b_{12} and b_{13} receive no analogous cross-term contribution. Cond. 2 at the observable level then translates to a relation on the substrate's direct second-order corrections:

$$b_{23}\text{\^{}}\mathbf{sub} = \mathbf{(4/3)}(b_{12}\text{\^{}}\mathbf{sub} + b_{13}\text{\^{}}\mathbf{sub}) + \mathbf{1/9}$$

The $+1/9$ augmentation is forced by $A_1's$ perturbative structure; deriving the substrate corrections that satisfy this from the cubic-lattice Yukawa is the quantitative open task. The ratio b_{23}^sub/(b_{12}sub+b_{13}^sub) at leading order in the

hierarchy m_e/m_τ takes the value $4/3 = 4(d-1)/(2d)$, and the sharp version of the question is whether a one-loop staggered-fermion calculation on the cubic lattice produces this specific ratio from the $C_2(T_1)$ Casimir structure.

We treat Condition 2 as a substrate-level relation and quote our angle predictions on the understanding that the predictions sit at the layered level: their structural form is forced by cubic-group representation theory and the Layer 1 closure above, but the specific 4/3 coefficient depends on a Layer 2(a)-tractable substrate computation that has not been completed. Empirical agreement at 0.07σ on $\sin^2\theta_{12}$ is, in this sense, a nontrivial test of the substrate's specific operator structure.

At the angle level, Condition 2 is equivalent to

$$2\Delta_{12} + \Delta_{23} = \mathbf{0}$$

which is testable directly against data (Sec. 4.3). The combination $b_{12} - b_{13}$ remains free; we identify it in Sec. 4.2 with δ_CP (the natural identification, since it is the only remaining $O(\lambda^2)$ free parameter that does not enter any of the three angles). The present analysis therefore predicts all three angles but not δ_CP.

3.5 Numerical predictions

Substituting $b_{23} = 1$, $b_{12} + b_{13} = 3/4$, and $\lambda^2 = 1/(2\pi^2)$:

- $\sin^2\theta_{12} = \mathbf{1/3 - 1/(4\pi^2) = 0.30798}$
- $\sin^2\theta_{13} = \mathbf{4/(18\pi^2) = 0.02252}$
- $\sin^2\theta_{23} = \mathbf{1/2 + 1/(2\pi^2) = 0.55066}$

The first formula for $\sin^2\theta_{13}$ follows from $A^2 = 2/3$ and $\lambda^2 = 1/(2\pi^2)$ together with the constraints on A_1; it does not use the normalization conditions on A_2. The other two additionally use Conditions 1 and 2.

4. The $\sin^2\theta_{12}$ prediction

4.1 Numerical value and comparison with data

The headline prediction is

$$\sin^2\theta_{12} = \mathbf{1/3 - 1/(4\pi^2) = 0.30798}$$

Comparison with current measurements:

Source	$\sin^2\theta_{12}$	σ from prediction
This work	0.30798	—
JUNO direct	0.3092 ± 0.0087	0.14σ
Post-JUNO global	0.3085 ± 0.0073	0.07σ

The prediction agrees with the post-JUNO global fit at 0.07σ. Both inputs entering the prediction are independently motivated: $1/3$ is the A_4 TBM zeroth-order value, and $\lambda^2 = 1/(2\pi^2)$ is the squared Cabibbo angle, fixed independently by the cubic Brillouin geometry (matching the observed $\lambda = 0.22508$ to 0.04%).

4.2 Comparison with column-preservation patterns

Tribimaximal mixing [1] was the leading phenomenological description of neutrino oscillation data until Daya Bay measured a nonzero θ_{13} [2]. Since then, modified TBM patterns have been the subject of a substantial literature [3, 4, 5, 6]. The two best-known modifications preserve one column of the original TBM matrix exactly: TM1 (first column) and TM2 (second column). Each gives a one-parameter family that accommodates the observed θ_{13} but predicts different values for the other two angles. Most existing TBM-modification proposals start from a discrete *flavor symmetry* — typically A_4, S_4, or a Z_2 residual — postulated as an axiom of the lepton sector. The minimal modifications then preserve one column of the TBM matrix or one residual symmetry of the neutrino mass matrix, yielding one-parameter correlations between angles. Post-JUNO, He [9] finds that TM2 is excluded at $>3.5\sigma$ while TM1 remains viable; Zhang [10] reports similar results.

Comparison summary:

Approach	Symmetry origin	$\sin^2\theta_{12}$	σ
TM1	postulated A_4 flavor	0.3184	1.35σ
TM2	postulated A_4 flavor	0.3408	4.4σ
TM3	postulated; $\theta_{13} = 0$	—	excluded
This work	**cubic group O of 3D space**	**0.3080**	**0.**07σ

(Comparing against post-JUNO global fit 0.3085 ± 0.0073. TM1 uses $\sin^2\theta_{12} = 1 - (2/3)/\cos^2\theta_{13}$; TM2 uses $\sin^2\theta_{12} = 1/(3\cos^2\theta_{13})$; both with $\sin^2\theta_{13} = 0.02195$.)

Three differences distinguish the present prediction from TM_n:

(i) Symmetry origin. TM1, TM2, and TM3 take a discrete flavor symmetry as an axiom of the lepton sector. The cubic-group prediction here uses $A_4 \subset$ O, where O is the symmetry group of three-dimensional space and the three generations transform as the T_1 irrep (Sec. 2.1; forced by the framework's Bravais-lattice uniqueness and anomaly-cancellation chain).

(ii) Independence of predictions. TM1 gives a one-parameter correlation, $\sin^2\theta_{12} = 1 - (2/3)/\cos^2\theta_{13}$, fixing θ_{12} given θ_{13} but predicting neither separately. The cubic-group analysis predicts all three angles as fixed numerical values, with no parameter common to them (Sec. 3.5).

(iii) CP violation. He's surviving pattern predicts sin δ_CP = ± 0.998, close to maximal. Here, δ_CP is identified with the only remaining free parameter at $O(\lambda^2)$, namely $b_{12} - b_{13}$, which does not enter any of the three angles. The angle predictions therefore stand independently of any value of δ_CP, and the present paper makes no prediction for it.

The bottom line: the cubic-group prediction is closer to the post-JUNO global fit by a factor of about 20 in σ-distance than the best surviving column-preservation pattern, and at the same time makes more falsifiable claims (three independent angles plus the sum rule below).

4.3 The sum-rule discriminator

The cubic-group prediction implies

$$\mathbf{2}\sin^2\theta_{12} + \sin^2\theta_{23} = \mathbf{7/6} = \mathbf{1.1667}$$

versus the observed $2(0.3085) + 0.561 = 1.178 \pm 0.020$ (using the post-JUNO $\sin^2\theta_{12}$ and the NuFIT 6.0 NO best fit for $\sin^2\theta_{23}$). The agreement is at 0.6σ. This sum rule is the angle-level statement of $b_{23} = (4/3)(b_{12} + b_{13})$, which is structural in the cubic-group setup. TM1 and TM2 do not produce this combination as an exact sum rule; in TM2, $\sin^2\theta_{12}$ is determined by θ_{13} alone with no θ_{23} correlation. A direct measurement of $2\sin^2\theta_{12} + \sin^2\theta_{23}$ at the 0.005 level is therefore the sharpest data-side test that distinguishes the two classes of TBM modification.

5. The remaining angles

5.1 $\sin^2\theta_{13}$

The formula for $\sin^2\theta_{13}$ is fully derived: at $O(\lambda^2)$ the A_2 parameters cancel from $|U_e3|^2$, leaving $\sin^2\theta_{13} = A^4\lambda^2 = 4/(18\pi^2) = 0.02252$. The NuFIT 6.0 best fit (normal ordering) is 0.02195 ± 0.00056, agreeing at 1.01σ. Equivalently,

$$\sin\theta_{13} = A^2\lambda = \mathbf{(2/3)} \cdot \mathbf{1/(\pi\sqrt{2})} = \mathbf{0.1500}$$

giving $\theta_{13} = 8.63°$, compared with the observed $8.56° \pm 0.10°$.

5.2 $\sin^2\theta_{23}$

The formula gives $\sin^2\theta_{23} = 1/2 + 1/(2\pi^2) = 0.5507$. NuFIT 6.0 (NO) gives $\sin^2\theta_{23} = 0.561^{+0.012}_{-0.015}$, agreeing at 0.74σ. The prediction lies in the second octant, in line with the current global preference. The sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ ties this directly to $\sin^2\theta_{12}$:

$$\sin^2\theta_{23} = \mathbf{7/6} - \mathbf{2}\sin^2\theta_{12}$$

A ± 0.0014 ultimate JUNO error on $\sin^2\theta_{12}$ then implies a ± 0.0028 error on the predicted $\sin^2\theta_{23}$, comparable to the current direct measurement.

5.3 Summary

Angle	Prediction	Measurement	σ
$\sin^2\theta_{12}$	$1/3 - 1/(4\pi^2) = 0.3080$	0.3085 ± 0.0073	0.07
$\sin^2\theta_{13}$	$4/(18\pi^2) = 0.02252$	0.02195 ± 0.00056	1.01
$\sin^2\theta_{23}$	$1/2 + 1/(2\pi^2) = 0.5507$	$0.561^{+0.012}_{-0.015}$	0.74

All three predictions agree with observation within 1.1σ.

6. Robustness

6.1 Cumulative empirical match

The 0.07σ match for $\sin^2\theta_{12}$ does not stand alone. It sits within a set of empirical matches all using the same cubic-group structural inputs:

Observable	Structural input	Prediction	Observed	Match
Cabibbo angle	Brillouin distance $\pi\sqrt{2}$; chirality $1/\lvert q\rvert$	$\lambda = 1/(\pi\sqrt{2}) = 0.22508$	0.22500 ± 0.00067	0.04%
Koide angle (charged leptons)	C_2/d^2 for cubic group	$\theta_0 = 2/9 = 0.22222$	0.222204 ± 0.000010	0.02%
Solar mixing $\sin^2\theta_{12}$ (this work)	A_4 representation $1/3$ + Cond 2 cubic relation	$1/3 - 1/(4\pi^2) = 0.30798$	0.3085 ± 0.0073	0.07σ
PMNS sum rule (this work)	Cond 2 cubic relation	$2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$	1.178 ± 0.020	0.6σ
Reactor mixing $\sin^2\theta_{13}$ (this work)	$A^2\lambda$ projection geometry	$4/(18\pi^2) = 0.02252$	0.02195 ± 0.00056	1.01σ
Atmospheric mixing $\sin^2\theta_{23}$ (this work)	Cond 2 + sum rule	$1/2 + 1/(2\pi^2) = 0.55066$	$0.561^{+0.012}_{-0.015}$	0.74σ

Two structural parameters appear repeatedly: $\lambda^2 = 1/(2\pi^2)$, the squared Cabibbo angle from the cubic Brillouin geometry, and $A^2 = 2/3$, the off-

democratic projection $\sin^2(\angle(\mathrm{e_i},\ \hat{h}))$ from the cubic-group representation. Both enter multiple observables at sub-percent or sub-σ precision. Under any reasonable prior over flavor models, this is non-trivial structural agreement: a single set of inputs produces six distinct empirical matches, each of which would have to be explained as coincidence in a flavor-symmetry framework that postulates the cubic structure rather than deriving it.

6.2 Sensitivity to Condition 2

The factor 4/3 in Condition 2 (Sec. 3.4) is the one substantive structural input. Its observable consequences enter the sum rule $2\Delta_{12} + \Delta_{23} = 0$ and, through Conditions 1 and 2 jointly, the absolute predictions for $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$.

A small shift to $4/3 \to (4/3)(1+\epsilon)$ propagates to $\Delta_{12} \to \Delta_{12} + (1/2)\cdot\epsilon\cdot\Delta_{23}$ at first order in ϵ. With $\Delta_{23} \approx 1/(2\pi^2) \approx 0.05$, a fractional shift $\epsilon = 0.10$ (a 10% shift in Cond 2's structural coefficient) gives a shift in $\sin^2\theta_{12}$ of ~0.0025 — comparable to JUNO's ultimate-precision uncertainty of ± 0.0014 and roughly half of JUNO's first-result uncertainty of ± 0.0087. The current 0.07σ agreement therefore tests Cond 2 at the few-percent level on its structural coefficient — a non-trivial test, since 4/3 is precisely $2A^2$ where $A^2 = 2/3$ is the projection factor independently fixed by the cubic representation geometry.

6.3 Status of Condition 2

Condition 2 admits the structural decomposition into two parts established in Sec. 3.4: an A_1 component closed at Layer 1 (the TBM sum rule is structurally protected against the A_1 perturbation by the $\cos^2\theta_{13}$ dressing alone, requiring no substrate input), and an A_2 component reducing to the sharp question why $b_{23}/(b_{12}+b_{13})$ takes the specific value $4/3 = 2A^2$ in the framework's bijection. The reactor angle $\sin^2\theta_{13}$ is independent of Condition 2 (the A_2 coefficients cancel from $|U_e3|^2$) and remains Layer 1 unconditional structural. The angle predictions $\sin^2\theta_{12}$ and $\sin^2\theta_{23}$ are layered: their structural form is forced by cubic-group representation theory and the Layer 1 closure of the A_1 component, while the specific 4/3 coefficient is a Layer 2(a)-tractable open question — sharply specified rather than a generic Layer 2(b) "alignment in T_2-space" — concerning a single numerical ratio with the structural reading $4(\mathrm{d}-1)/(2\mathrm{d})$ for $\mathrm{d} = 3$.

Positive support for the Layer 2 classification. Beyond the structural decomposition, an explicit substrate calculation in [18, §7.3] confirms that Condition 2 is not satisfied by the natural form of the substrate Yukawa structure: for the natural cubic-symmetric staggered Yukawa structure on T_1 BZ corners, both the leading and subleading channels produce equal off-diagonal $Y^{(2)}$ entries by cubic symmetry, failing Condition 2 by a hierarchy factor ~17. The framework's bijection therefore has structure beyond the natural cubic-symmetric form — Condition 2's empirical match indicates the bijection introduces direction-dependent corrections that the natural form lacks. The 0.07σ

match is therefore a test of the bijection's specific T_2-channel alignment.

Cubic-group consistency test. The PMNS sum rule via Condition 2 is one observable in a six-prediction simultaneous-match cluster derived from the structural identity $A^2 = C_2/d$ on the d=3 cubic lattice. The other five are Wolfenstein A (1.2%), $\sin^2\theta_{13}$ (0.4%), Koide θ_0 (0.02%), m_u/m_d (0.6%), and |V_cb| (0.4σ). All six match observation at sub-percent or sub-σ precision simultaneously [18, §7.7]. This places the sum rule's 0.6σ match in a broader pattern of cubic-group structural consistency across the framework.

Discretization-determined Layer 2. Condition 2 is therefore not a free parameter to be fit, but a derivable consequence of the lattice's discretization scheme. The bijection space in [18] is constrained by linear dynamics, T-invariance, P-indivisibility, and cycle ergodicity, leaving a discretization-scheme freedom (field quantization, lattice size, timestep, boundary conditions, rounding rule) rather than abstract permutation freedom. The specific $4/3 = 2A^2$ coefficient required by Condition 2 is *discretization-determined*, not an independent Wilson coefficient. The Layer 2(a) promotion path is concrete: a one-loop staggered-fermion calculation on the cubic lattice that produces the $b_{23}/(b_{12}+b_{13}) = 4/3$ ratio from the $C_2(T_1)$ Casimir structure would close Condition 2 to Layer 2(a). The 0.07σ empirical match at $\sin^2\theta_{12}$ and 0.6σ at the sum rule are tests of this discretization scheme — the predictions follow from the bijection's substrate structure rather than from independently adjusted Wilson coefficients.

The sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ is the most directly testable consequence of Condition 2; its empirical match at 0.6σ tests the bijection's alignment property at the few-percent level on the structural coefficient. The discriminating power of the sum rule against TM1, TM2, and TM3 column-preservation patterns (Sec. 4.2, 4.3) is independent of the Layer 2(a) vs 2(b) status of Condition 2; the empirical match is what matters for the discrimination.

7. Falsifiability

The prediction is sharp: $\lambda^2 = 1/(2\pi^2)$ is fixed independently by the Cabibbo angle, and the coefficient $1/(4\pi^2)$ in Δ_{12} has no adjustable input. Three experimental directions test it.

Sub-percent precision on $\sin^2\theta_{12}$. Over its 30-year design lifetime, JUNO is projected to reach $\sin^2\theta_{12} = 0.3092 \pm 0.0014$, a 6× improvement over the current first result. At that precision the prediction 0.30798 is tested at 0.86σ (using the JUNO 59-day central value) or 0.36σ (using the post-JUNO global central value). An ultimate JUNO 2σ disagreement with 0.30798 would falsify the prediction.

The sum rule. The prediction $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$ is sharper than either angle alone: it ties JUNO's reactor measurement to the atmospheric measurements at DUNE, T2HK, and Hyper-Kamiokande. As both improve,

the sum rule will be tested at the 0.005 level — a sharper test than either side separately.

δ_CP. The free parameter $b_{12} - b_{13}$ of A_2 does not enter any angle prediction, so the framework is consistent with any δ_CP. Long-baseline measurements at DUNE and Hyper-Kamiokande will determine this remaining parameter directly.

8. Conclusion

We have derived $\sin^2\theta_{12} = 1/3 - 1/(4\pi^2) = 0.3080$ from cubic-group flavor structure, matching the post-JUNO global fit at 0.07σ with no parameter fit to $\sin^2\theta_{12}$. The same construction predicts $\sin^2\theta_{13}$ and $\sin^2\theta_{23}$ within 1.1σ of the NuFIT 6.0 best fits [14], and forces the exact sum rule $2\sin^2\theta_{12} + \sin^2\theta_{23} = 7/6$, which distinguishes the prediction from the TM1, TM2, and TM3 column-preservation patterns. The reactor angle $\sin^2\theta_{13} = 4/(18\pi^2)$ is unconditional structural (Layer 1 in [18, §8.3]), based on the representation-theoretic identity $A^2 = (\mathrm{d}-1)/\mathrm{d} = C_2(T_1)/d$ for vector reps. The solar and atmospheric angle predictions are layered: their structural form is forced by cubic-group representation theory and the Layer 1 closure of the A_1 component (Sec. 3.4), while the specific 4/3 coefficient in Condition 2 ($b_{23} = (4/3)(b_{12}+b_{13})$) is a Layer 2(a)-tractable open question — sharply specified as the substrate ratio $b_{23}{}^{\mathrm{sub}/(b_{12}}\mathrm{sub}+b_{13}\hat{}\mathrm{sub})$ at leading order in the m_e/m_τ hierarchy. The empirical match at 0.07σ on $\sin^2\theta_{12}$ and at 0.6σ on the sum rule indicates the framework's bijection has the alignment property required by Condition 2; the predictions are therefore tests of the bijection's specific structure as much as of the underlying cubic-group framework. The Cabibbo scale $\lambda^2 = 1/(2\pi^2)$ entering the prediction is set independently by cubic Brillouin-zone geometry and matches the observed Cabibbo angle to 0.04%.

JUNO's projected long-term precision will test the prediction at sub-percent level over the next decade. Combined with atmospheric measurements at DUNE, T2HK, and Hyper-Kamiokande, this will sharply test the sum rule. A confirmed result at JUNO's ultimate precision would support cubic-group flavor structure as a viable origin of the lepton mixing pattern.

Acknowledgments

The author thanks the JUNO Collaboration for the rapid public release of their first oscillation results, and F. Capozzi, E. Lisi, F. Marcone, A. Marrone, and A. Palazzo for the prompt post-JUNO global fit that enabled the present comparison. Discussions with the broader emergent-quantum-mechanics community have informed the framework's positioning relative to existing approaches; specific intellectual debts are recorded in the references and in the underlying framework documentation [18].

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